

The Art of Guard

Last month we barely scratched the surface of the SELinux Targeted Policy that's shipped with RHEL 5. This month we'll try to delve deeper into its building blocks.



It has been an interesting journey for me ever since we started on "The Art of the Guard" series.

I never imagined the kind of response it would elicit and I must thank the readers for their responses to the content and their queries. One request has been to include illustrations in the text since "...a picture paints a thousand words." I shall try to accommodate as many suggestions as I can. Unfortunately, this article also does not carry any illustrations, but I am positive that the subsequent ones will.

In the previous article we started exploring the Red Hat Targeted Policy as shipped with Red Hat Enterprise Linux 5. In Dan Walsh's words: "The Targeted Policy goal is to lock down all processes that listen for network connections and pretty much

all processes that start at boot. Processes that are started by a logged in user were unconfined (*unconfined_t*). Services started by the *init* scripts that did not have a policy were also run in an unconfined domain (*initrc_t*)."

We also listed the users, roles and types that are inbuilt by default in the Red Hat Targeted Policy. Now, we shall explore some other building blocks of the Targeted Policy.

Booleans

Booleans, as we all know, are variables that can either be set as true or false. Booleans enhance the effect of SELinux policies by letting the system administrator fine tune a policy. A policy may protect a certain daemon or service by applying various access control rules. In real world scenarios, a system administrator would not like to implement all the access controls specified in the policy.

This is where Booleans help. Booleans create conditional access controls based on their value. As an example, the *httpd* (Apache Web Server) subject has the following Booleans in the targeted policy:

```
allow_httpd_mod_auth_pam
allow_httpd_bugzilla_script_anon_write
httpd_enable_ftp_server
allow_httpd_squid_script_anon_write
allow_httpd_anon_write
httpd_can_network_relay
httpd_disable_trans
httpd_tty_comm
httpd_unified
httpd_rotatelogd_disable_trans
httpd_builtin_scripting
httpd_enable_cgi
allow_httpd_nagios_script_anon_write
```

